



THE AMERICAN UNIVERSITY OF KURDISTAN
THE 10TH ANNIVERSARY

College of Arts and Science
Master of Science in Artificial Intelligence (AI)
The American University of Kurdistan

Final Exam Assignment: Artificial Intelligent
Spring-2026

AI-Based Smart Cardiopulmonary Blood Flow Control System

Submission of project

- Code: Fully developed, clear, and well-commented code must be submitted.
- Short report (max 5 pages).

Basic Information

- A cardiopulmonary bypass (CPB) system is a medical device used during surgery.
- It temporarily replaces the function of the heart and lungs. The system:
 - pumps blood (like the heart)
 - adds oxygen to blood (like the lungs)
- A rotary blood pump is responsible for controlling the blood flow to maintain stable and safe blood flow to the patient.
- Maintaining a stable blood flow (e.g., ~4 - 6 L/min) is critical for patient safety.

Project Details

To control blood flow, a simple AI based fuzzy logic controller is implemented in a cardiopulmonary bypass rotary blood pump. The controller is designed to maintain a constant blood flow of 6 liters per minute by adjusting the speed of the rotary pump. The system uses two input variables: the pressure difference (ΔP) measured between the inlet and outlet ports of the pump, and the reference blood flow setpoint. The output variable is the pump speed (N). The blood flow (Q), supplied to the patient, ranges from 0 to 10 L/min. The linguistic variables for flow are Low, Medium, and High, and their membership functions are shown in Figure 1.

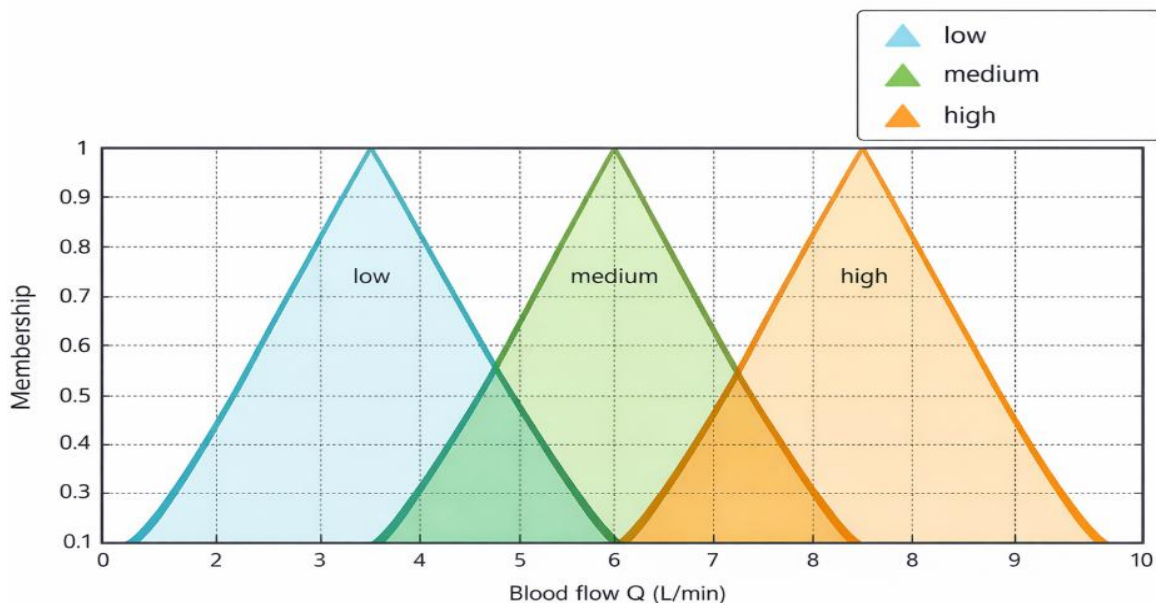


Figure 1. Linguistic variables for flow and membership functions.

The pressure difference (ΔP) ranges from 50 to 150 mmHg. The linguistic variables for ΔP are Low, Medium, and High, with corresponding membership functions shown in Figure 2.

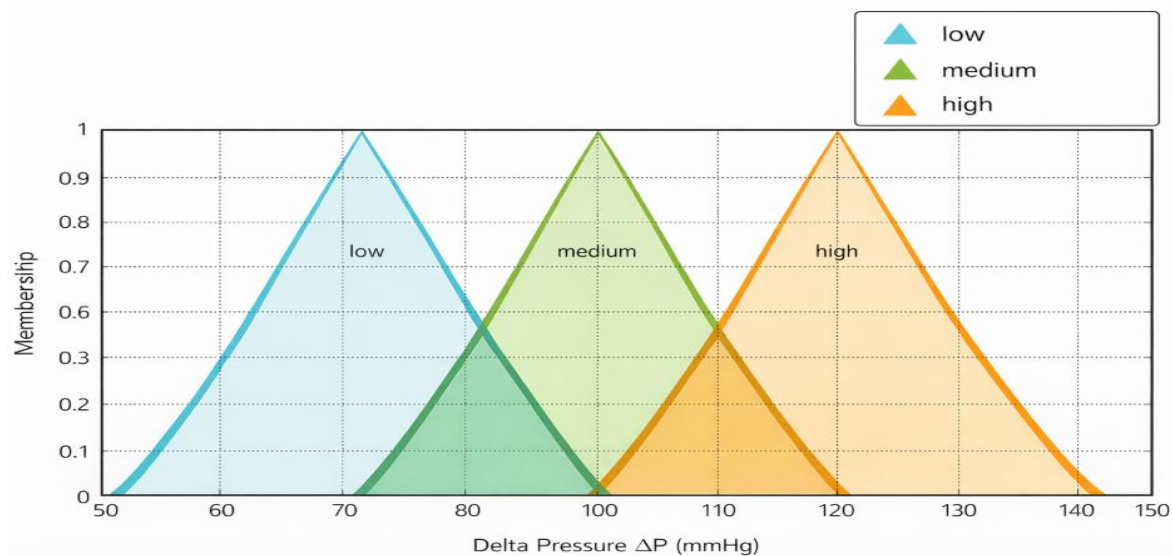


Figure 2. linguistic variables for ΔP with membership functions

The motor speed (N) determines the power delivered by the pump, thereby controlling the blood flow. The speed ranges from 2800 to 3200 rpm, with linguistic variables Slow, Medium, and Fast, as shown in Figure 3.

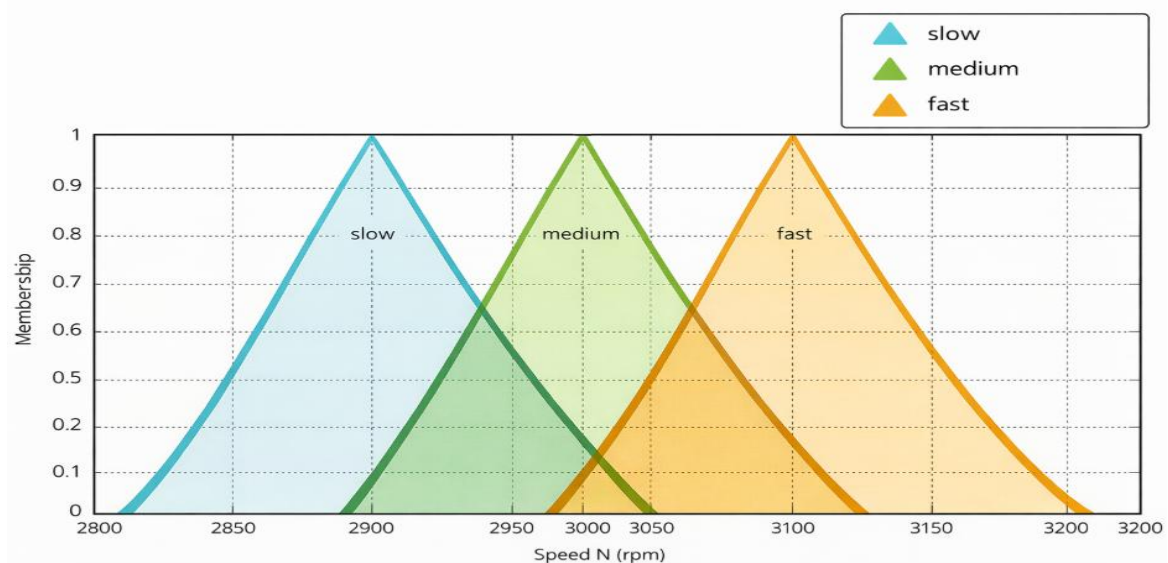


Figure 2. linguistic variables for motor speed with membership functions

The universe of discourse for all variables and their ranges are summarized in Table 1.

Table 1. Universe of discourse

Linguistic variables	Range
Flow (Q)	0 – 10 L/min
Delta pressure (ΔP)	50 – 150 mmHg
Pump velocity (N)	2800 – 3200 rpm

Table 2 presents the fuzzy IF - THEN rules used to adjust the pump speed (N) based on the measured pressure difference (ΔP) and blood flow (Q).

Table 2. AI-Based Fuzzy Rule Base for Rotary Blood Pump Control

$\Delta P \setminus Q$	Low Flow	Medium Flow	High Flow
Low ΔP	Fast	Medium	Slow
Med ΔP	Medium	Medium	Slow
High ΔP	Medium	Medium	Slow

Notes:

Convert the table into rules, for example: IF ΔP is low AND Q is low THEN Speed = fast

Tasks:

- 1- Apply the fuzzy rules shown in Table 2 and draw the aggregated output fuzzy set for the output variable pump speed (N) when the fuzzy system is given:
 - delta pressure, $\Delta P = 84$ mmHg
 - blood flow rate, $Q = 6$ L/min
- 2- Using the centroid defuzzification method, determine the crisp speed of the rotary pump when:
 - $\Delta P = 84$ mmHg
 - $Q = 6$ L/min

***** End of Assignment*****